

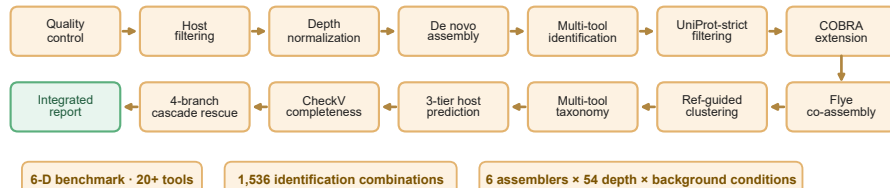
1 · Public Data Acquisition*public_metadata_pipeline*

- NGDC GSA + NCBI SRA retrieval
- AI-driven metadata curation
- Literature-traced completion
- Parallel download (aria2c · prefetch)
- 6 SCI-grade landscape plots

Desktop GUI — manual editing & visualization

2 · Benchmark-Driven Discovery Pipeline*virome_discovery_pipeline*

13 stages · every default parameter benchmark-validated

**3 · Rapid Known-Virus Analysis***virome_analysis_pipeline*

- Immediate detection
- Evolutionary characterization
- Quantification (TPM · RPM · depth)
- Co-infection & association
- Variant detection
- Pan-virome landscape

4 · Semi-automatic GenBank Submission*virome_submission_pipeline*

- Topology triage · MIUVIG metadata templates
- Sequin / suvtk packaging → .sqn

Dedicated desktop GUI

CROSS-CUTTING DESIGN STRATEGIES**Data Reduction****99.8%** *host-filtering efficiency*

- Reference-guided pre-clustering
- Host-based sequence partitioning

Quality Enhancement**4-branch** *cascade genome rescue*

- COBRA resolves de Bruijn breakpoints
- Flye co-assembly merges cross-sample
- BLASTN → PlantVirusDB rebuild
- bwa-mem2 + viral_consensus / ragtag
- Genus-length rescue as final fallback

Reliability Enhancement**F1 = 0.942** *identification (UniProt-strict)*

- UniProt-strict ID ≈ 5-tool ensemble
- Family accuracy 94.1% (+72.3 pp)
- 3-tier host + plant-specific filter